

Software Requirements Specification

for

Food delivery System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed description of the Food Delivery System. This Software Requirements Specification (SRS) describes the functional and non-functional requirements of the system intended to provide users with a single application to browse restaurants, order food items, make payments, and track deliveries.

This document serves as a reference for developers, designers, testers, and stakeholders to understand the system's behavior and constraints.

1.2 Document Conventions

This SRS document follows standard Software Engineering documentation conventions.

- *Section headings and subheadings are numbered hierarchically for clarity.*
- *Functional and non-functional requirements are described using clear and concise statements.*
- *Mandatory requirements are expressed using the word “**shall**”, while optional features use “**may**”.*
- *All technical terms and abbreviations are explained when first introduced.*
- *Higher-level requirements apply to detailed requirements unless explicitly stated otherwise. No special fonts or colors are used, except headings for readability.*

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- **Developers:** *To understand system functionality and implementation requirements.*
- **Project Managers:** *To plan development schedules and allocate resources.*
- **Testers:** *To derive test cases and validation criteria.*
- **End Users:** *To gain a high-level understanding of system features.*
- **Documentation Writers:** *To prepare user manuals and help guides.*

*Readers are advised to begin with **Section 1 (Introduction)** to gain an overview of the system.*

- *Developers and testers should focus on the functional and non-functional requirements sections.*
- *Project managers should review the product scope and constraints.*
- *End users may refer to the system overview and feature descriptions.*

1.4 Product Scope

The Food Delivery System is a software application designed to manage the complete food ordering and delivery process through a single platform. The system enables users to browse restaurants, select food items, place orders, make online payments, and track delivery status until the food is successfully received.

Benefits:

- *Eliminates long queues and waiting times.*
- *Avoids continuous phone calls for ordering and order status updates.*
- *Saves time and effort for both customers and restaurant staff.*

Objectives:

- Provide an online interface for food ordering.
- Allow users to place food orders anytime and from anywhere.
- Improve efficiency and accuracy in food delivery operations.

The system supports the goal of digital transformation by offering a convenient and scalable solution for food delivery services.

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Pressman, Roger S., Software Engineering: A Practitioner's Approach, McGraw-Hill Education.
- REST API Design Guidelines, Open Web Standards Documentation.
- Android Developer Documentation, Google Developers.
- Web application UI/UX design standards and guidelines.

2. Overall Description

2.1 Product Perspective

The Food Delivery System is a **new, self-contained software product** developed to digitalize and automate the complete food ordering and delivery process. It is not a replacement for an existing system but a standalone application designed to integrate multiple services such as restaurant management, online payments, and delivery tracking into a single platform.

The system functions as part of a larger ecosystem that includes:

- User mobile and web applications
- Restaurant management systems
- Payment gateway services
- Location and mapping services

The application interfaces with external systems such as **payment gateways, map and location services, and notification services** through Application Programming Interfaces (APIs). These integrations enable secure payments, real-time tracking, and communication between users, restaurants, and delivery personnel.

2.2 Product Functions

At a high level, the Food Delivery System provides the following major functions:

- User registration and authentication
- Browsing restaurants and food items
- Food cart management and order placement
- Online payment processing
- Nearest restaurant discovery based on user location
- Order confirmation and notification handling
- Real-time delivery tracking
- Order status updates until successful delivery

Detailed functional requirements are described in Section 3 of this document.

2.3 User Classes and Characteristics

The system identifies the following user classes:

1. **Customers (End Users)**
 - Use the application to browse menus, place orders, make payments, and track deliveries.
 - Require a simple and user-friendly interface.
 - May have minimal technical knowledge.
2. **Restaurant Staff / Restaurant Admin**
 - Manage menus, accept or reject orders, and update order status.
 - Require access to order and payment information.
 - Have moderate technical expertise.
3. **Delivery Personnel**
 - Receive delivery requests and update delivery status.
 - Use location tracking features.
 - Require real-time navigation and notification support.
4. **System Administrator**
 - Manages users, restaurants, and system configurations.
 - Has full access privileges and high technical expertise.

Customers are considered the **primary user class**, while other classes support system operations.

2.4 Operating Environment

The Food Delivery System will operate in the following environment:

- **Hardware Platform:** Smartphones, tablets, laptops, and desktop computers
- **Operating Systems:**
 - Android Operating System (OS)
 - iOS Operating System (OS)
 - Web browsers on Windows, Linux, and macOS
- **Server Environment:** Cloud-based servers
- **Supporting Software:**
 - Web browsers (Google Chrome, Mozilla Firefox, etc.)
 - Map and location service APIs
 - Payment gateway services

2.5 Design and Implementation Constraints

The development of the Food Delivery System is subject to the following constraints:

- Compliance with data security and privacy standards
- Use of secure password hashing mechanisms
- Integration with third-party payment gateways
- Dependence on location services for distance calculation and tracking
- Support for high traffic through scalable system design
- Use of REST (Representational State Transfer) APIs for communication
- Reliable internet connectivity required for real-time updates
- Programming and design standards must ensure maintainability and performance

2.6 User Documentation

The following user documentation will be delivered along with the software:

- User manual for customers
- Restaurant management guide
- Delivery personnel usage guide
- Online help and FAQs
- Installation and setup documentation (if applicable)

All documentation will be provided in **digital format** and written in simple, easy-to-understand language.

2.7 Assumptions and Dependencies

NA

3. External Interface Requirements

3.1 User Interfaces

The Food Delivery System shall provide a **graphical user interface (GUI)** that is simple, responsive, and easy to navigate for all user classes. The user interface will be designed following standard usability principles to ensure smooth interaction.

- The application shall provide separate interfaces for **customers, restaurant staff, delivery personnel, and system administrators**.
- Common interface components such as **login, logout, navigation menu, and help options** shall be available across all screens.
- The user interface shall display clear and consistent error messages for invalid inputs or system failures.
- Buttons, icons, and input fields shall follow consistent layout and labeling standards.
- The interface shall support touch-based interaction for mobile devices and mouse/keyboard interaction for web users.
- Screen layouts shall be responsive to adapt to different screen sizes and resolutions.
- Confirmation messages shall be displayed for actions such as order placement, payment success, and delivery completion.

Detailed user interface designs will be documented separately in the User Interface Specification document.

3.2 Hardware Interfaces

The Food Delivery System shall interact with the following hardware components:

- **Mobile Devices:** Smartphones and tablets used by customers, delivery personnel, and restaurant staff.
- **Desktop and Laptop Computers:** Used for web-based access by administrators and restaurant staff.
- **GPS Hardware:** Used for location tracking and distance calculation.

The software shall access hardware features such as GPS through operating system-level services and APIs. All hardware interactions shall occur over standard communication protocols supported by the respective operating systems.

3.3 Software Interfaces

The Food Delivery System shall interface with the following software components:

- **Operating Systems:** Android OS, iOS OS, and web browser environments.
- **Database Management System:** Used to store user details, orders, payments, and restaurant data.
- **Payment Gateway Software:** For secure online transactions and payment status updates.
- **Map and Location Services:** For location mapping, distance calculation, and delivery tracking.
- **Notification Services:** For sending order and delivery status updates.

Data exchanged between software components shall include user credentials, order details, payment information, and location data. Communication between components shall be performed using **REST APIs** with structured data formats such as JSON.

3.4 Communications Interfaces

The Food Delivery System shall support network-based communication using standard protocols.

- Communication between client applications and servers shall use **HTTPS**.
- All sensitive data transmissions shall be encrypted to ensure security and privacy.
- The system shall support real-time updates for order status and delivery tracking.
- Notification messages shall be delivered using internet-based communication services.
- Data synchronization shall be handled to ensure consistency between user, restaurant, and delivery modules.

The system shall comply with standard communication and security practices to ensure reliable and secure data exchange.

4. System Features

This section describes the major system features of the Food Delivery System. Each feature represents a key service provided by the system and includes its description, priority, stimulus/response sequence, and functional requirements.

4.1 User Registration and Authentication

4.1.1 Description and Priority

This feature allows users to create an account and securely log in to the Food Delivery System using their credentials.

Priority: High

4.1.2 Stimulus/Response Sequences

- User opens the application.
- User selects the registration or login option.
- User enters required credentials (email and password).
- System validates the input data.

- System authenticates the user and grants access, or displays an error message if validation fails.

4.1.3 Functional Requirements

- **REQ-1:** The system shall allow users to register using email, password, and full name.
- **REQ-2:** The system shall validate user credentials during login.
- **REQ-3:** The system shall securely store passwords using hashing techniques.
- **REQ-4:** The system shall display appropriate error messages for invalid login attempts.

4.2 Restaurant and Food Item Browsing

4.2.1 Description and Priority

This feature enables users to browse available restaurants and view food items offered by each restaurant.

Priority: High

4.2.2 Stimulus/Response Sequences

- User logs into the system.
- User selects the browse restaurants option.
- System displays a list of nearby restaurants.
- User selects a restaurant.
- System displays the menu and food item details.

4.2.3 Functional Requirements

- **REQ-5:** The system shall display a list of available restaurants.
- **REQ-6:** The system shall show food items with price and description.
- **REQ-7:** The system shall allow users to search or filter food items.

4.3 Cart Management and Order Placement

4.3.1 Description and Priority

This feature allows users to add food items to a cart, modify the cart, and place an order.

Priority: High

4.3.2 Stimulus/Response Sequences

- User selects a food item.
- User adds the item to the cart.
- User modifies quantity or removes items if required.
- User confirms the order.
- System calculates the total cost and proceeds to payment.

4.3.3 Functional Requirements

- **REQ-8:** The system shall allow users to add or remove food items from the cart.
- **REQ-9:** The system shall display the total order cost dynamically.

- **REQ-10:** The system shall allow order modification before confirmation.

4.4 Payment Processing

4.4.1 Description and Priority

This feature enables users to make secure online payments for their orders.

Priority: High

4.4.2 Stimulus/Response Sequences

- *User selects a payment method.*
- *System redirects to the payment gateway.*
- *User completes payment.*
- *System receives payment status confirmation.*
- *System updates order status accordingly.*

4.4.3 Functional Requirements

- **REQ-11:** The system shall support online payment through payment gateways.
- **REQ-12:** The system shall confirm payment success or failure.
- **REQ-13:** The system shall update payment status using webhooks.

4.5 Order Tracking and Delivery Management

4.5.1 Description and Priority

This feature allows users to track their food delivery in real time until successful completion.

Priority: Medium

4.5.2 Stimulus/Response Sequences

- *User places an order.*
- *System assigns delivery personnel.*
- *Delivery personnel updates delivery status.*
- *System displays real-time delivery tracking.*
- *User confirms food received.*

4.5.3 Functional Requirements

- **REQ-14:** The system shall provide real-time delivery tracking.
- **REQ-15:** The system shall notify users of delivery status changes.
- **REQ-16:** The system shall confirm successful food delivery.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The Food Delivery System shall meet the following performance requirements to ensure efficient and reliable operation:

- *The system shall respond to user actions such as login, restaurant browsing, and order placement within **2–3 seconds** under normal network conditions.*
- *Payment processing responses shall be received within a reasonable time frame as defined by the payment gateway service.*
- *The system shall support multiple concurrent users without performance degradation.*
- *Real-time delivery tracking updates shall refresh periodically to provide accurate location information.*
- *The system shall handle peak usage periods, such as meal times, without system failure.*

These requirements ensure smooth user experience and timely order processing.

5.2 Safety Requirements

The Food Delivery System shall be designed to minimize risks of data loss, service disruption, or misuse:

- *The system shall prevent accidental or unauthorized modification of order and payment data.*
- *Transaction failures shall be handled gracefully without causing incorrect order placement or payment duplication.*
- *User data shall not be lost due to system crashes or unexpected shutdowns.*
- *Backup and recovery mechanisms shall be implemented to restore critical data if failures occur.*

The system shall follow general safety practices applicable to online service platforms.

5.3 Security Requirements

The Food Delivery System shall ensure data security and user privacy through the following measures:

- *The system shall authenticate users before granting access to protected features.*
- *User passwords shall be stored using secure hashing techniques.*
- *Sensitive data such as payment information shall be transmitted using encrypted communication channels.*
- *Role-based access control shall restrict system functions based on user roles.*
- *The system shall prevent unauthorized access, data tampering, and information leakage.*

The system shall comply with applicable data protection and security policies.

5.4 Software Quality Attributes

The Food Delivery System shall exhibit the following quality attributes:

- **Usability:** *The system shall be easy to use with minimal learning effort.*
- **Reliability:** *The system shall operate consistently without frequent failures.*
- **Scalability:** *The system shall support increasing numbers of users and orders.*
- **Maintainability:** *The system shall be easy to update and maintain.*
- **Portability:** *The system shall run on multiple platforms including mobile and web.*

- **Availability:** The system shall be accessible at all times except during scheduled maintenance.
- **Robustness:** The system shall handle invalid inputs and unexpected situations gracefully. Ease of use and reliability are given higher priority over other quality attributes.

5.5 Business Rules

The Food Delivery System shall operate under the following business rules:

- Only registered users shall be allowed to place food orders.
- Restaurants shall manage only their own menus and orders.
- Delivery personnel shall update only delivery-related information.
- System administrators shall have full access to manage users and system configurations.
- Orders cannot be modified after payment confirmation.

These business rules define operational boundaries and ensure controlled system usage.

6. Other Requirements

6.1 Database Requirements

The system shall use a database to store user profiles, restaurant details, food items, orders, payment status, and delivery information. The database shall ensure data consistency, integrity, and secure access.

6.2 Legal and Regulatory Requirements

The system shall comply with applicable data protection and privacy regulations. User data shall be collected and processed only for system functionality.

Appendix A: Glossary

Term / Acronym	Definition
SRS (Software Requirements Specification)	A document that describes the requirements of a software system
GUI (Graphical User Interface)	Visual interface through which users interact with the system
API (Application Programming Interface)	Interface that allows communication between software components
GPS (Global Positioning System)	Technology used for location tracking
REST (Representational State Transfer)	Architectural style for web-based services
HTTPS (Hypertext Transfer Protocol Secure)	Secure version of HTTP used for encrypted communication

Appendix B: Analysis Models

NA

Appendix C: To Be Determined List

NA